

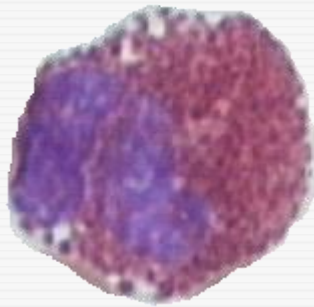
Eosinophilia

Prof.dr. Michèle van Vugt

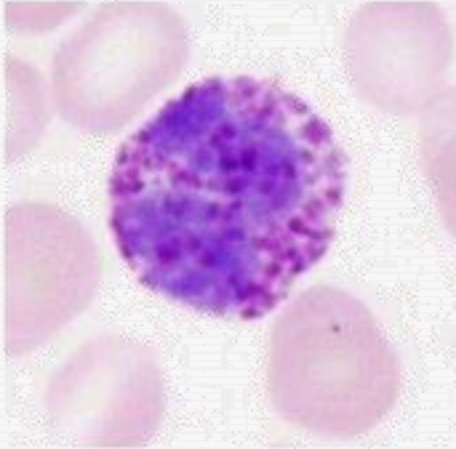
Amsterdam University Medical Center

m.vanvugt@amsterdamumc.nl

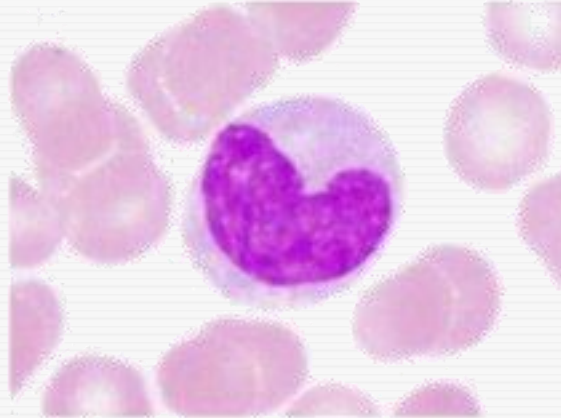
Eos, Aurora



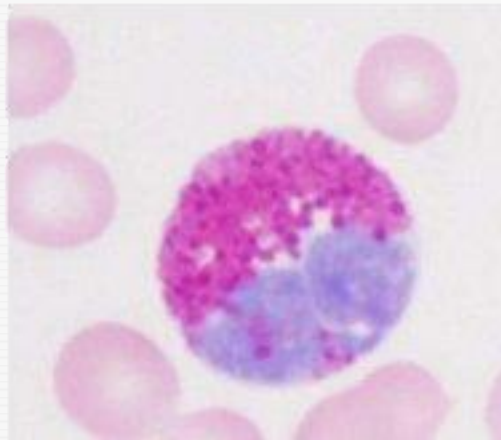
Basophil



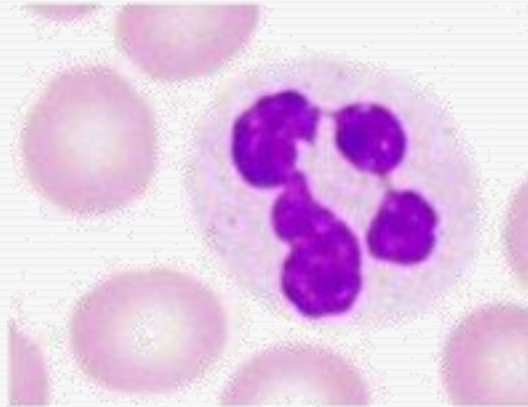
Monocyte



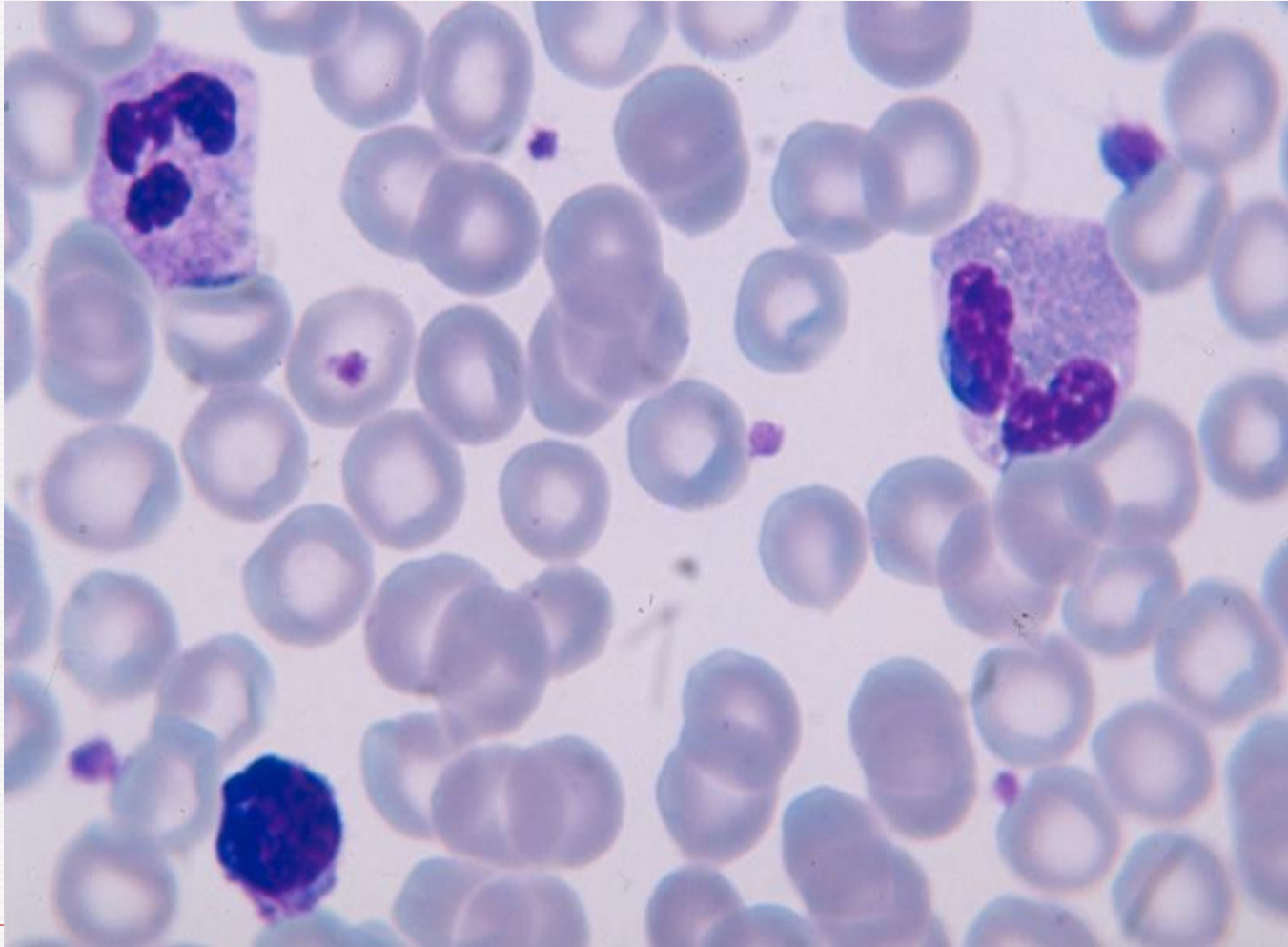
Eosinophil



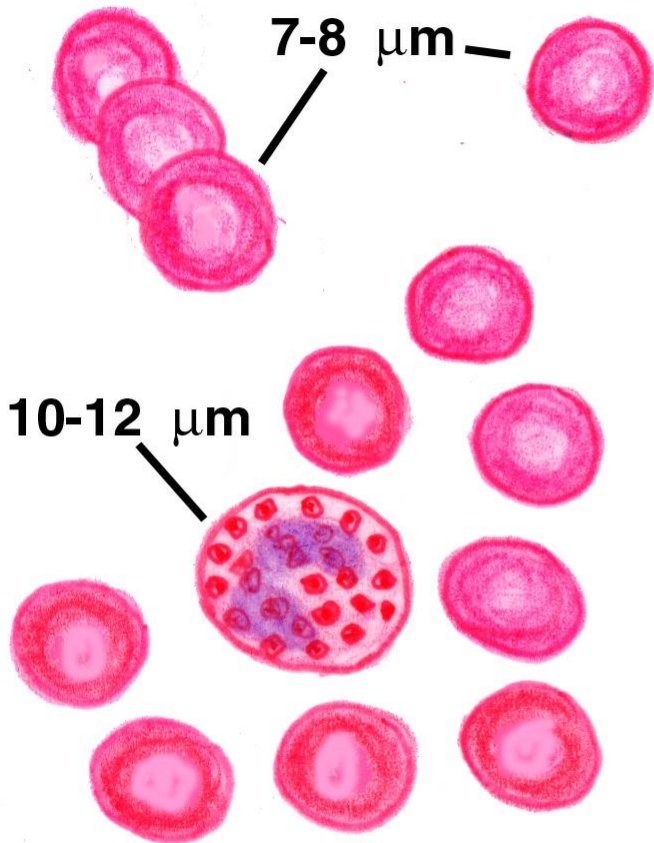
Neutrophil



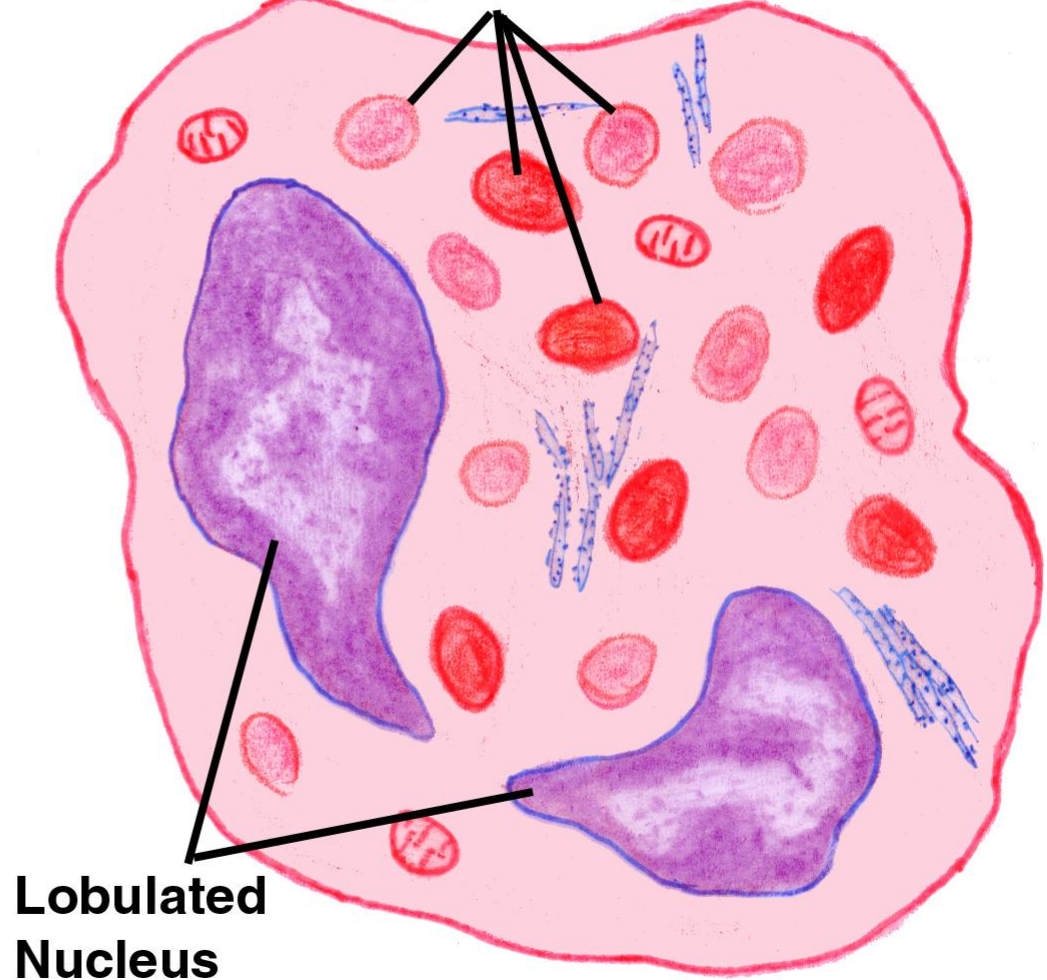
Eosinophils



Size relation to erythrocytes



Specific granules (lysosomes)



Eosinophils

- 1879 Ehrlich
- Hemopoietic stemcells bone marrow
- blood → tissues (1:100-500)
- $\pm$ 20 granules
 - peroxidase and lysosomal enzymes

Eosinophils

- cytotoxic to parasites but also to normal mammalian cells
- induces histamine release from basophils and mastcells
- Increase in eosinophil numbers is T-cell dependent
- 3 cytokines promote:
 - IL-3
 - IL-5
 - GM-CSF

Eosinophils

- Eosinophils express membrane receptors
 - adhesion receptors
 - Fc receptors
 - mediator receptors

and produce

- chemotoxins and potent mediators like
 - platelet activating factor
 - leukotrienes
 - lipid mediators

Eosinophil function

- “dampen down” histamine dependent reactions
- “ameliorate inflammatory responses”
- kill helminths (IgG, IgE, Complement)

Eosinophil function

Normal count:

- $400 / \mu\text{l} = 0.4 \times 10^9/\text{l} = 400 \times 10^6/\text{l}$
note: 765 healthy students: 15 – 650 μl

Eosinophil numbers vary with:

- age (young, higher: <1 yr. up to 700; <10 yr. up to 550)
- time of the day (night/early morning high; late morning low)
- dependent corticosteroids/ ACTH
- exercise
- environmental stimuli

Counting of eosinophils

- | | accuracy |
|-------------------------|----------|
| • wet count “chamber” | 2. |
| • differential count | 3. |
| • automated cell counts | 1. |

Definition of eosinophilia

- mild 400 - 1500 • $10^6/L$
- moderate 1500 - 5000
- marked ≥ 5000

Question: Causes eosinophilia ?

Question: Causes eosinophilia ?

Allergy

Infectious Diseases

Malignancies

Auto-immune Diseases

Question: Causes

Allergy

- atopia
- allergy-medicaments

Infectious Diseases

- helminths

Malignancies

- metastatic cancer, Hodgkin lymphoma
- hypereosinophilic syndrome

Auto-immune Diseases

- allergic en granulomatous vasculitis (Church-Strauss)
- polyarteriitis nodosa
- reumatoide arthritis
- eosinophilic fasciitis
- skindiseases as pemphigus vulgaris, dermatitis herpetiformis

Classification of eosinophilia

Familial (rare, autosomal dominant)

Acquired

Primary (males >> female, clonal)

- Clonal
- Idiopathic

Secondary

- Tissue-invasive **helminth infection**
- Bacterial or viral infections (rare)
- **Drugs/ Toxins**
- **Allergy**
- Autoimmune
- Malignancy (metastatic cancer, Hodgkin lymphoma)
- Endocrinopathies

Mrs. R

- Mrs. R. 65 years, Hindustan Surinam background
- In NL since 1983
- Dec 2004 – Jan 2005 Surinam

Mrs. R

- Feb 2005: emergency room AMC:
- Since Surinam / return:
- Diarrhoea, Nausea/vomitus, Anorexia
- Weight loss, Wheezing, cough

Mrs. R

Medical history:

- 1989 *Fasciola hepatica* infection
- 1994 Ovary adenoma (testosterone producing)
- TIA
- Reflux oesophagitis,
- Episodes of cough
- Polyposis nasi
- Hypertension

Mrs. R

Medication (adherence?):

- Carbasalaatcalcium
- Nifedipin
- Captopril
- Omeprazol
- Salmeterol/fluticason inhaler

Mrs. R

Physical examination

- BP: 204/124 mmHg, PR: 94 / min
- Wheezing
- Tender upper abdomen

Mrs. R

Laboratory examination

WBC: $16.1 \bullet 10^9/\text{L}$

Diff: 38% neutros

48 %	eos
-------------	------------

$\sim 7700. 10^6/\text{L}$

12 % lympho' s

2 % mono' s



What kind of laboratory tests?

1. PCR
2. Serology for exotic infections
3. Parasitological investigation for amoebiasis
4. Christa punction

Differential diagnosis?

Differential diagnosis?

Helminth infection

Allergy

Something different

Helminth infections ??

Helminth infections

Strongyloidiasis

Filariasis

Schistosomiasis

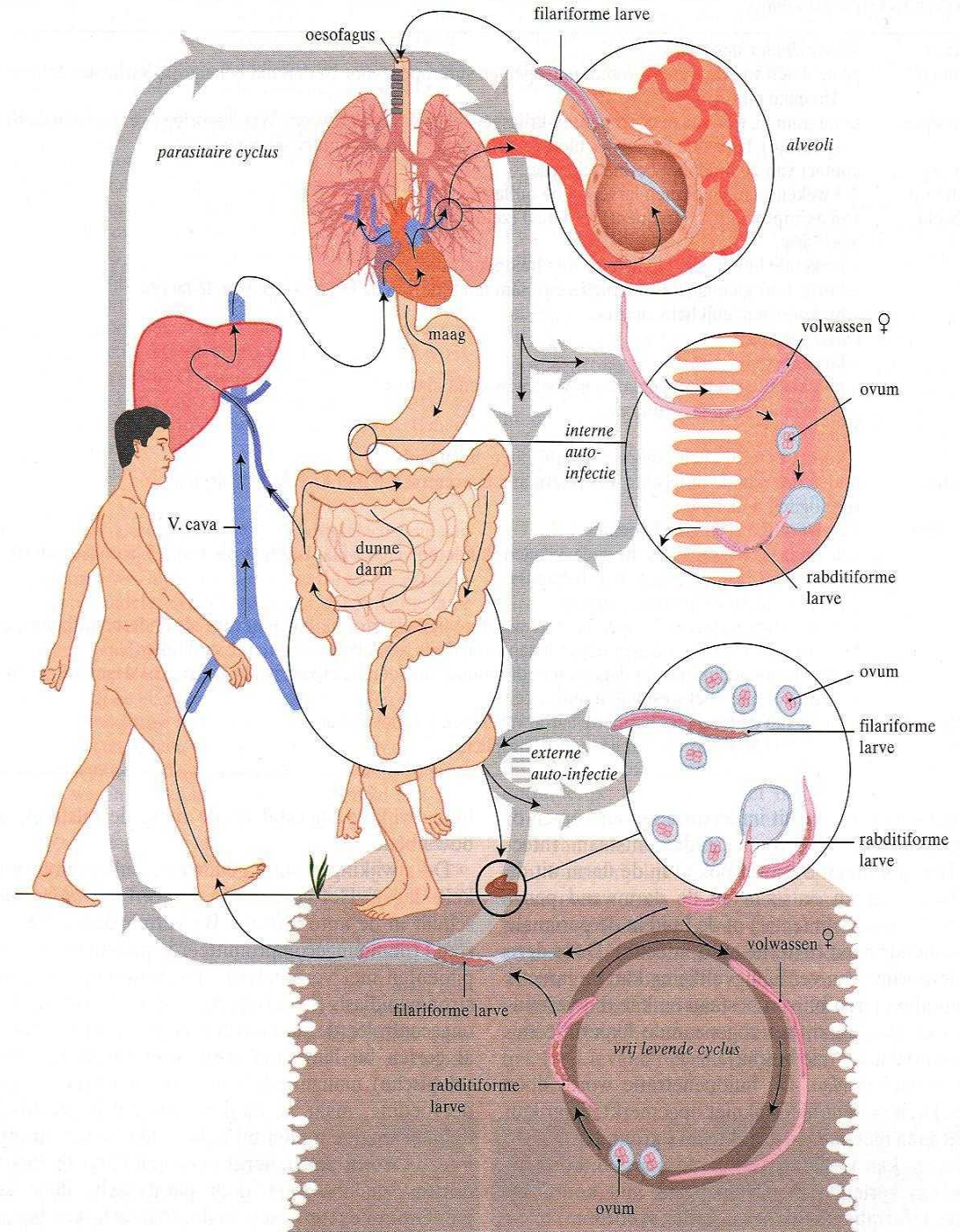
Löffler syndrome

Fascioliasis

Strongyloides stercoralis

- Warm wet (sub) tropical areas
- Transmission by contact with soil or close contacts with an infected person

Strongyloides cycle



Strongyloidiasis

- Incubation: 2-3 weeks (to larvae in stool)
- From asymptomatic to fatal

Strongyloides, larva currens



Strongyloides, larva currens



S. stercoralis larva



Strongyloidiasis

- asymptomatic
- larva currens
- urticarial rash
- cough, shortness of breath, wheezing,
- pulmonary infiltrates (Löffler's syndrome)
- abdominal problems:
 - epigastric pain, indigestion
 - watery diarrhoea, cramping pain

Strongyloides, hyperinfection

- immunocompromised patients (many causes)
- fever, enteritis, obstruction, sepsis, shock
- pulmonary infiltrates, meningitis
- peritonitis, endocarditis

Strongyloides, diagnosis

- larvae in stool (Baermann test; culturing; agar plate test)/duodenum
- In hyperinfection, larvae everywhere (blood, urine, sputum, BAL-fluid, CSF, gastric fluid, peritoneal fluid)
- serology

Strongyloides, treatment

- ivermectin 200 microg/kg, once
- albendazole 400 mg, bid, 3d

Albendazol



Ivermectine



Thiabendazol withdrawn



Strongyloides hyperinfection and immunocompromised patients

- ivermectin 200microg/ kg od at least 3 d,
“sec. prophylaxis”: 1 dose every fortnight/ or
month
- albendazol 400 mg bid, 14 d,
and then 1 dose every fortnight or month?

Mrs. R

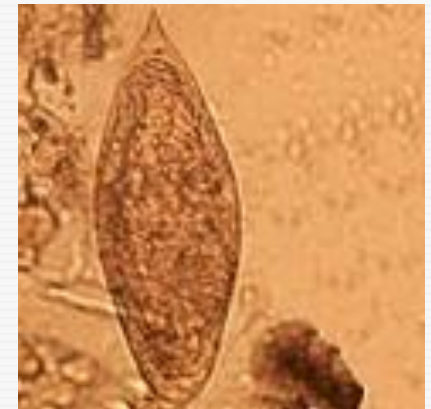
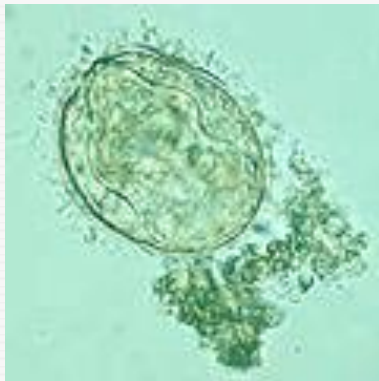
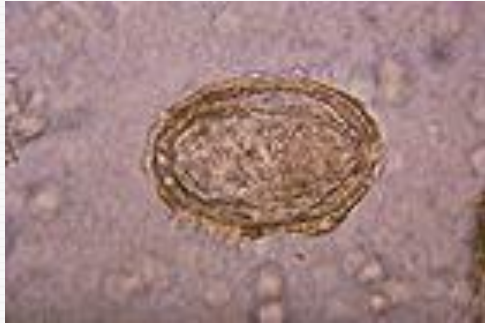
Laboratory examination (cont.)

Stool: no Strongyloides

Serology: negative

Schistosomiasis

- *Schistosoma mansoni*
- *S. intercalatum*
- *S. haematobium*
- *S. japonicum*
- *S. mekongi and malayi*



S. Haematobium

S. Mansoni

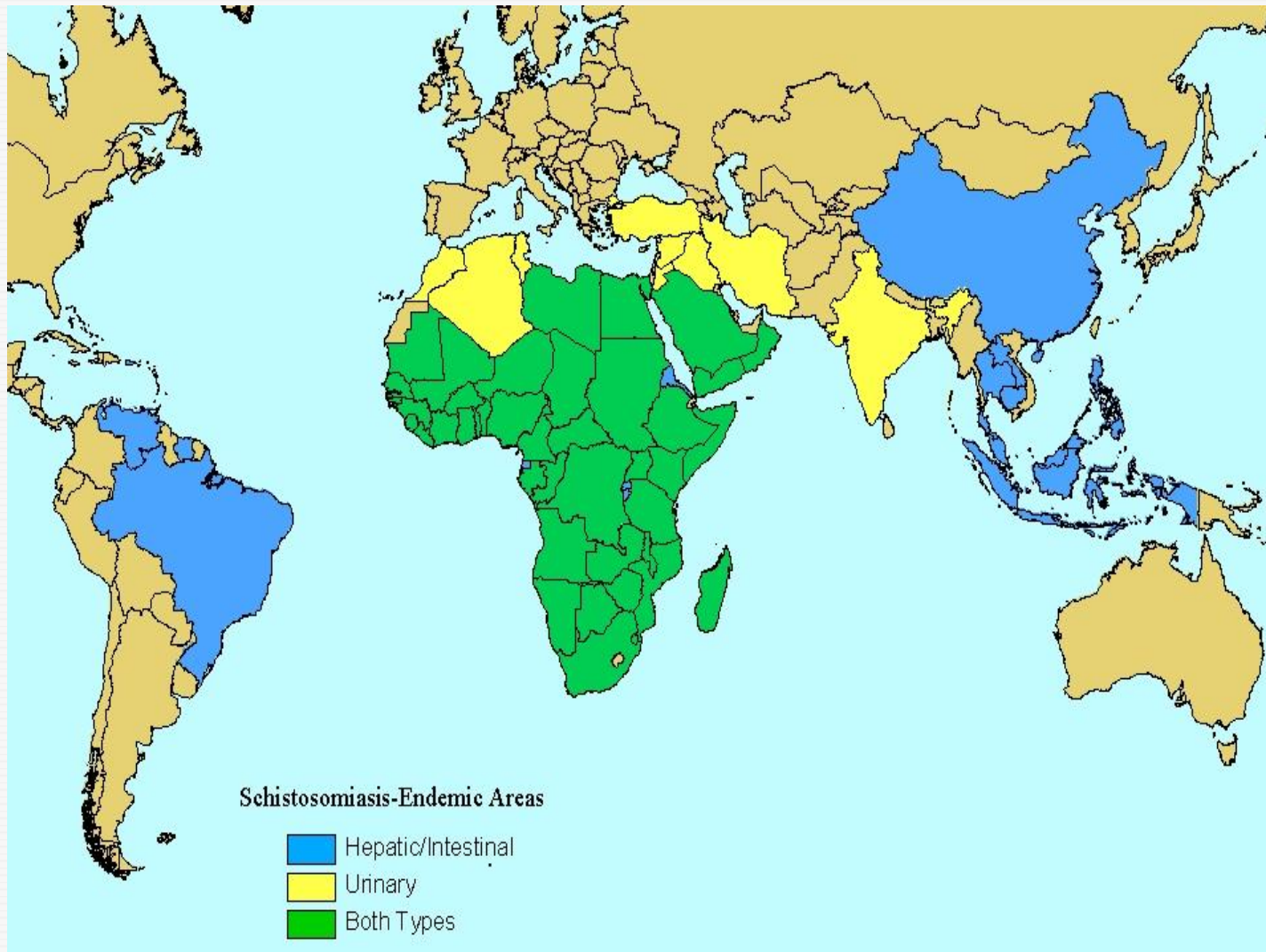
S. Mekongii

Trematoda

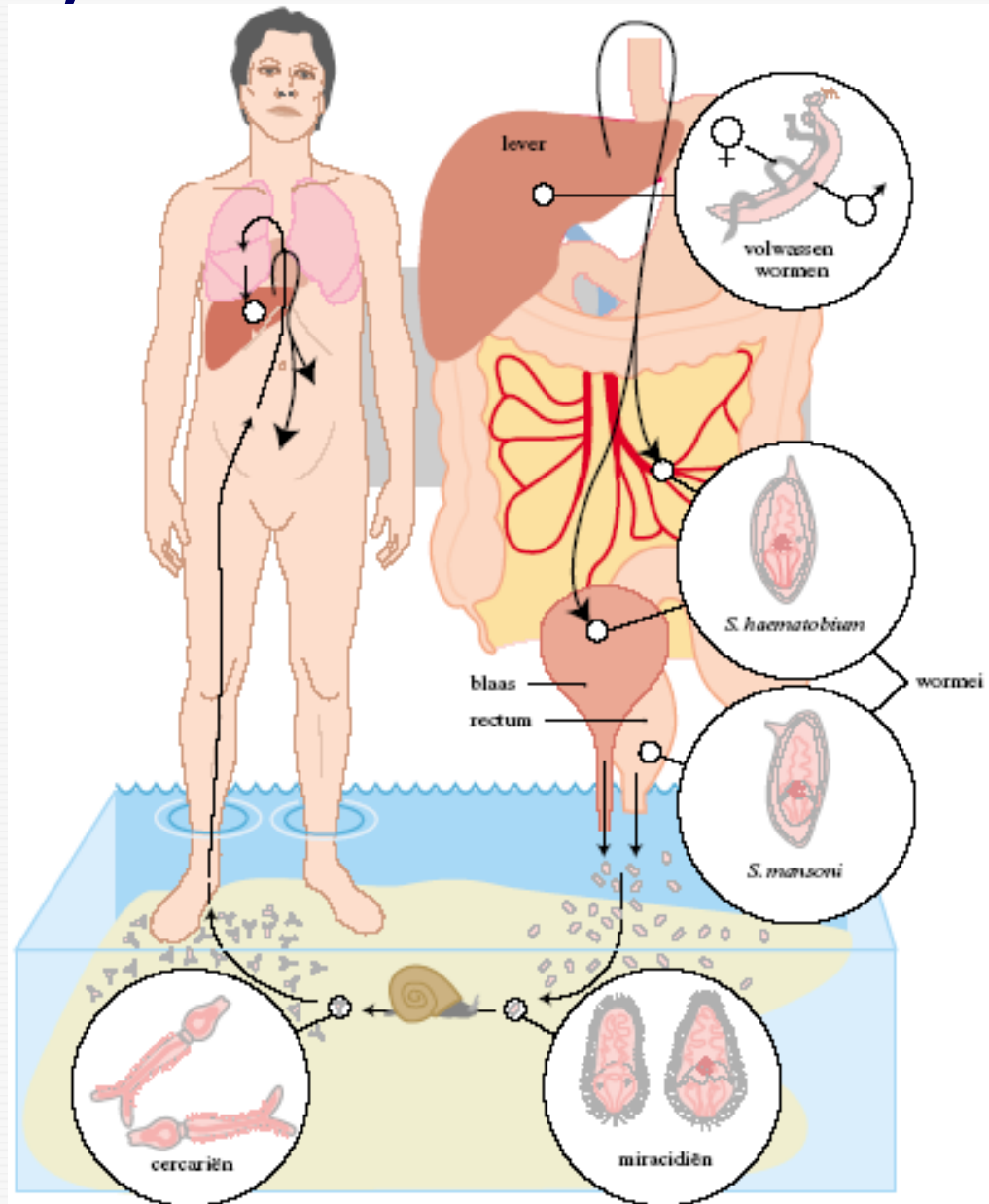
S. Japonicum

S. Intercalatum

Schistosomiasis, geographic distribution



Schistosomiasis, cycle



Schistosomiasis, acute ~ Katayama fever

- 4-7 weeks after exposure (~ first egg release)
- fever (may be prolonged for weeks, intermittent)
- malaise
- abdominal pain, diarrhoea
- cough, wheezing
- urticaria
- oedema
- lymphadenopathy
- eosinophilia

S. mansoni, egg

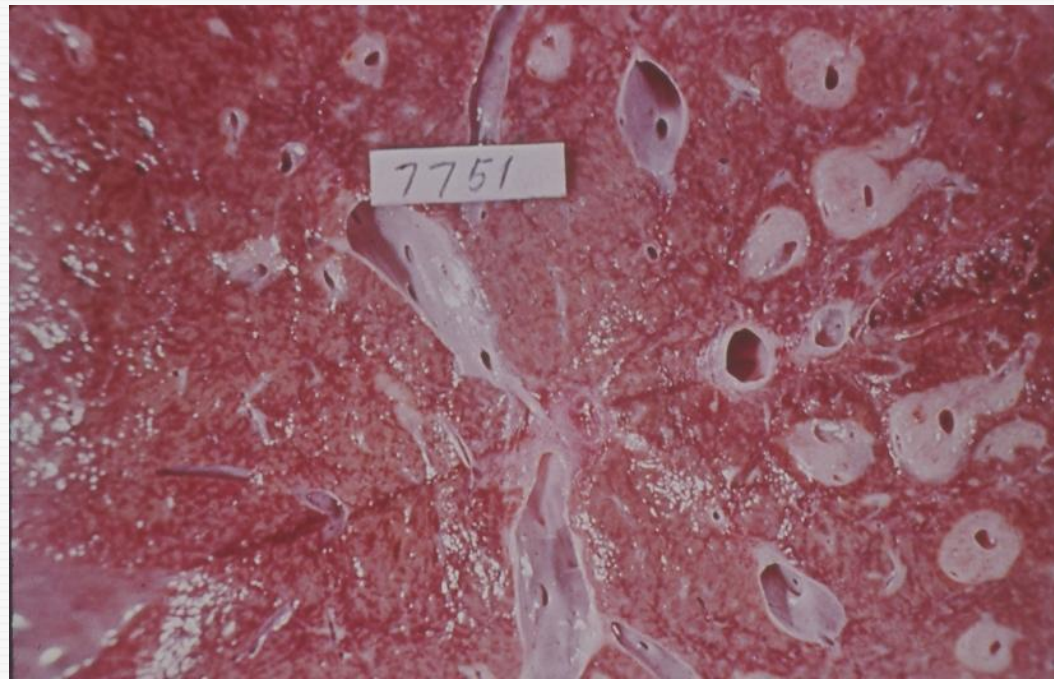
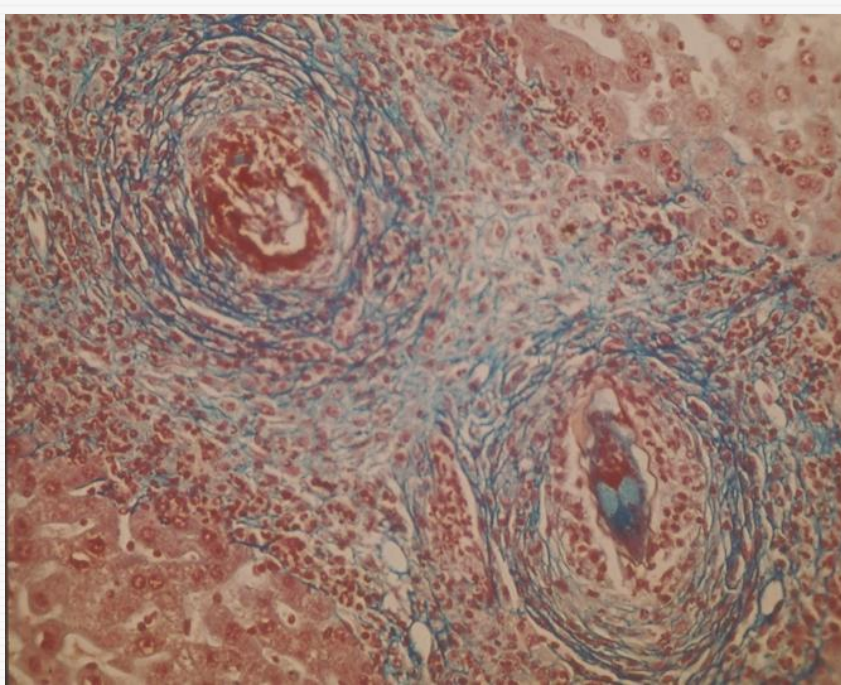


S. haematobium, egg



Schistosomiasis, chronic

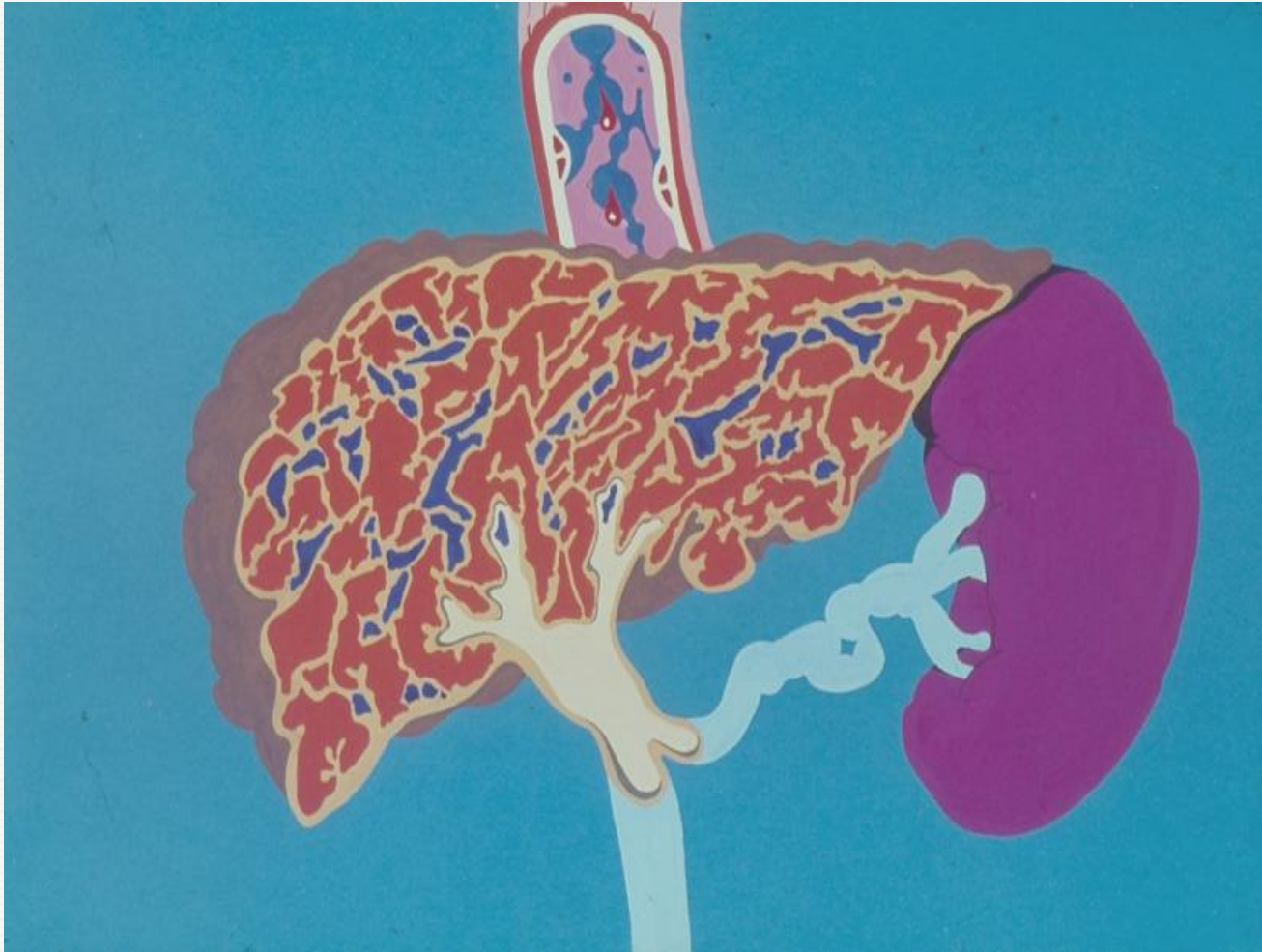
- caused by eggs and granulomatous reaction to eggs.
- little eosinophilia





Hepatosplenic schistosomiasis

Pre-sinusoidal portal hypertension



S.haematobium, bladder, hydroureter



S. haematobium, hydroureter,
hydronephrosis, calcified bladder



Neurological complications of schistosomiasis



Schistosomiasis, diagnosis

- eggs stool, urine (rectal, bladder biopsy)
- serology antigen and antibody detection
- eosinophilia (Acute: high; chronic: mild/N)
- haematuria, proteinuria
- ultrasound, X-ray, CT scan, MRI

Schistosomiasis, treatment

Praziquantel

- *S.mansoni* & *S. haematobium*: 40 mg/kg once
- *S.japonicum*: 2 x 30 mg/kg on 1 day
- CNS: praziquantel + corticosteroid

Treatment

- Praziquantel 40 mg/kg single dose
- *S. japonicum* 2 x 30 mg/kg
- CNS: praziquantel + corticosteroid
- Katayama: + corticoids

Mrs. R

Laboratory examination (cont.)

Stool: no *Schistosoma* eggs

Urine: no eggs

Serology: negative

Filarial infections

- Filariae
- threadlike nematodes
- adults in lymphatics or connective tissue
- vector transmitted
- may live for decades
- females produce large numbers of larvae

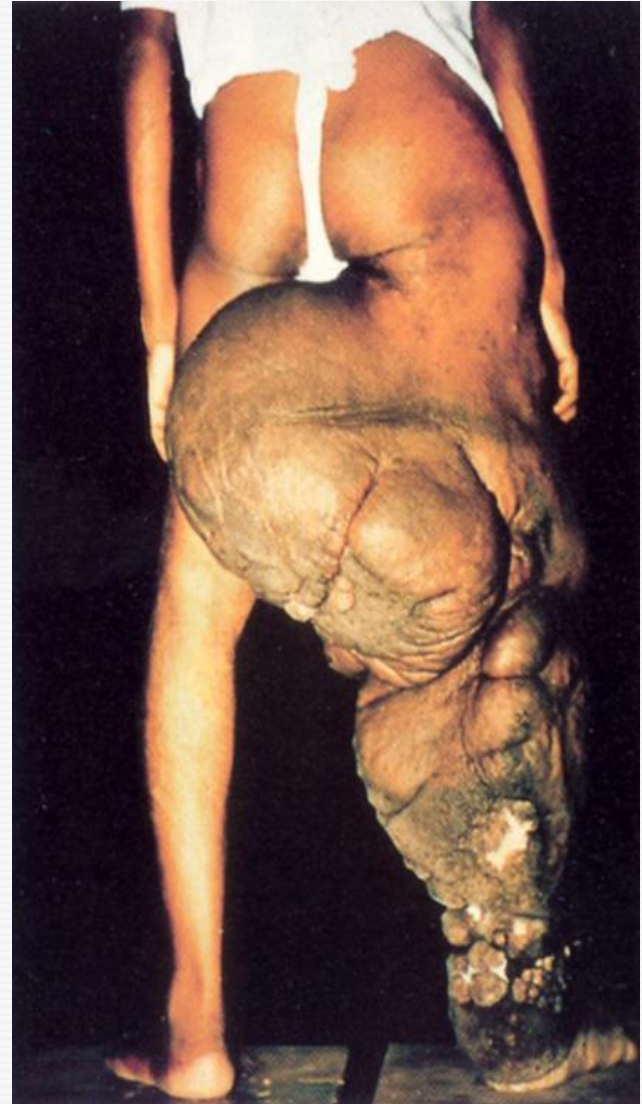
Filarial infections, different diseases

- Filariasis, lymphatic filariasis
 - *Wuchereria bancrofti*: cosmopolitan,
 - esp. India, China, Indonesia
 - *Brugia malayi*: SE-Asia, Indonesia, India
 - (*Brugia timori* SE Indonesia)

Filarial infections, different diseases

- *Loa loa*
- *Onchocerca volvulus* (river blindness)
- *Mansonella* spp.
- *Dirofilaria* spp.

Lymphatic filariasis, elephantiasis



Filarial infections, diagnosis

- microfilariae in blood, urine (ascites; hydrocele)
- serology

Filarial infections, treatment

- Diethylcarbamazine (DEC),
(microfilariae & adults)
- Ivermectin (microfilariae)
- Antibiotics (doxycycline)

Tropical pulmonary eosinophilia

- ~ *Wuchereria bancrofti* or *Brugia malayi*
- most cases from India
- alveolitis because of trapping of microfilariae and adults in the lung
- granulomas

Tropical pulmonary eosinophilia

- male: female 4:1
- fever, non-productive cough
wheezing especially at night
- malaise, fatigue, loss of weight
- marked eosinophilia
- serology positive
- no microfilariae
- X ray: diffuse small interstitial,
reticulo-nodular infiltrates

TPE, treatment = diagnosis

- treatment
- DEC* 3 dd 2 mg/kg, opbouwen!
- 10-21 d. ~ total dose 72 mg/kg
- + corticoids

*Diethylcarbamazine

Mrs. R

Laboratory examination (cont.)

Filaria a.b.: ++

Blood: no microfilariae

Mrs. R, DEC treatment

Little effect

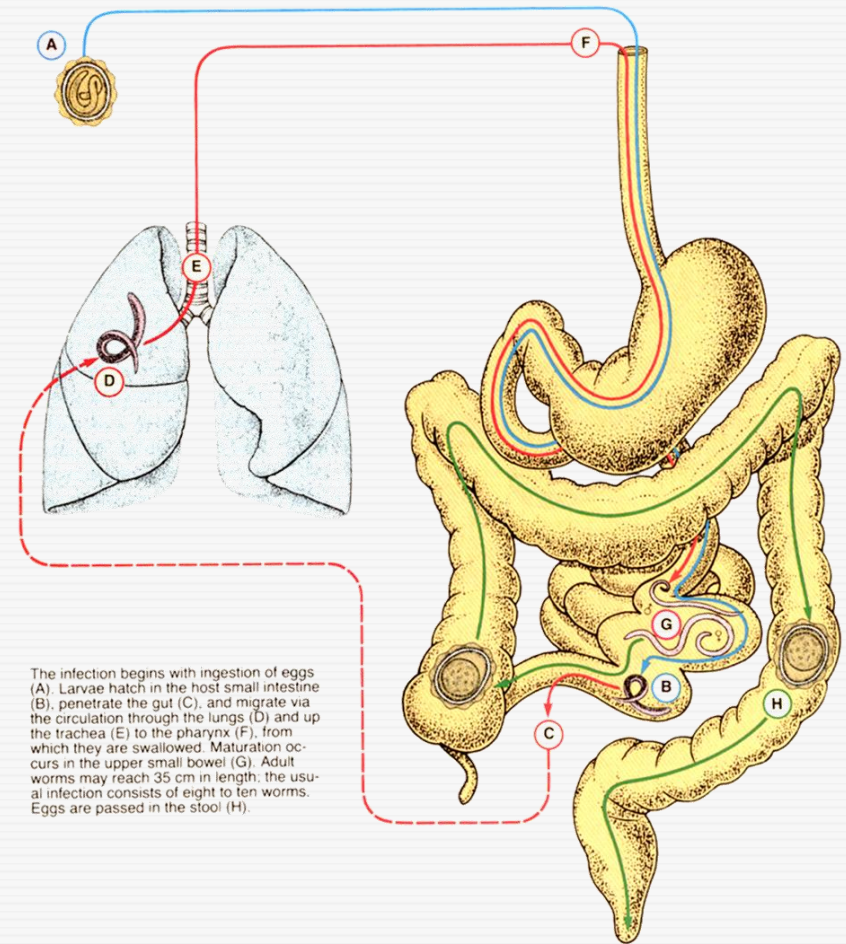
Löffler syndrome

- Acute eosinophilic pneumonia
- migrating infiltrates
- minimal clinical manifestations

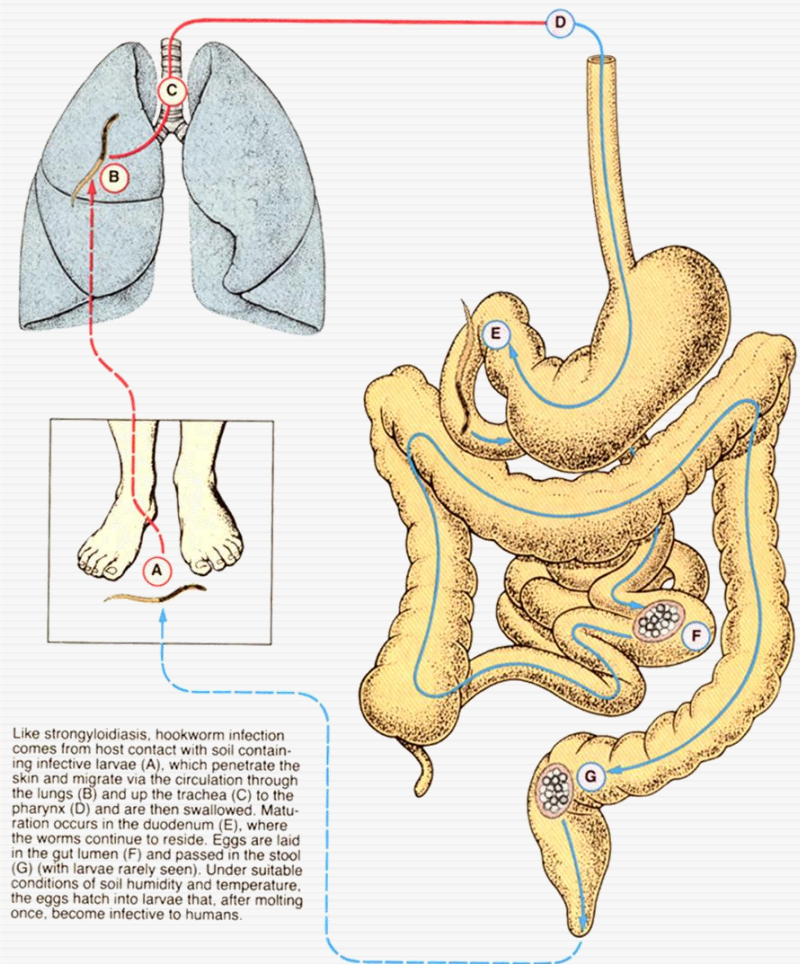
DD

- other acute / chronic eos. pneumonias
- Churg Strauss angiitis

Ascariasis, initial tissue migration



Hookworms, initial tissue migration



Mrs. R

Laboratory examination (cont.)

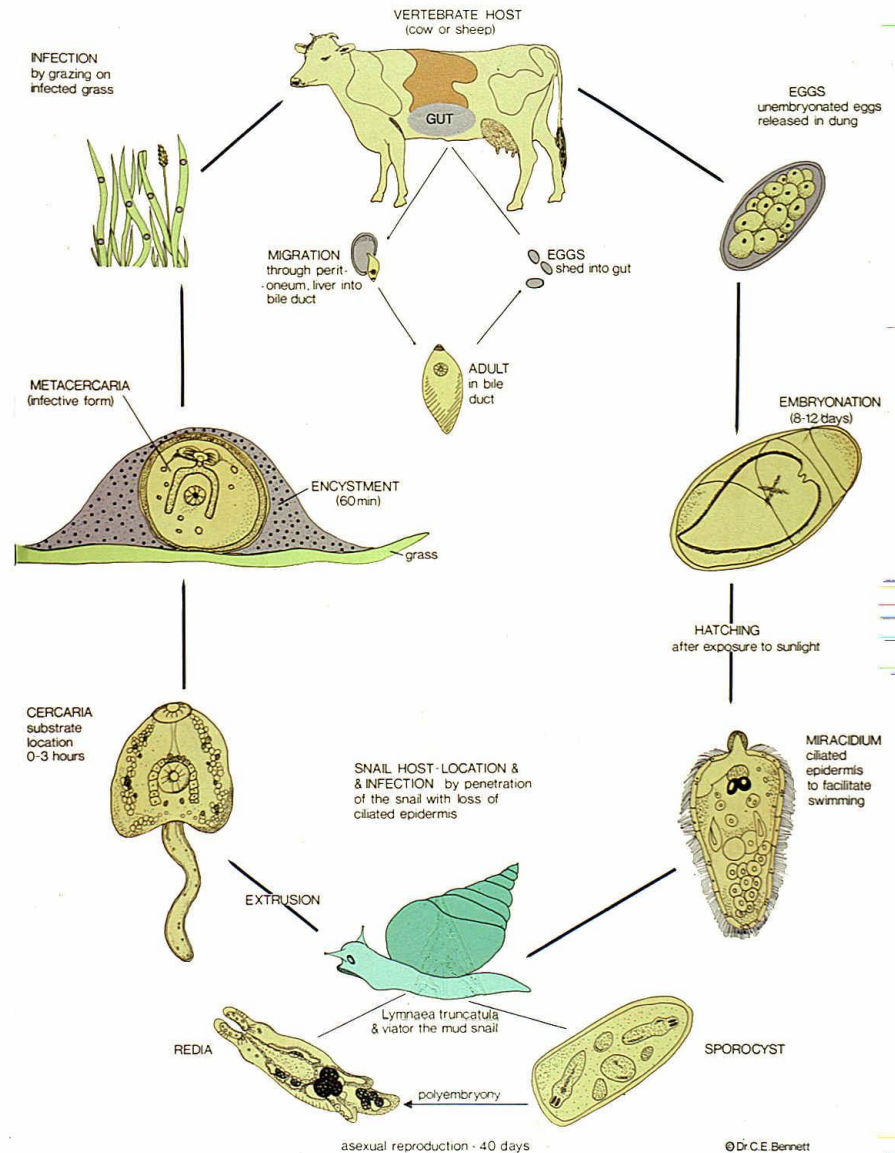
Stool parasitology: no eggs

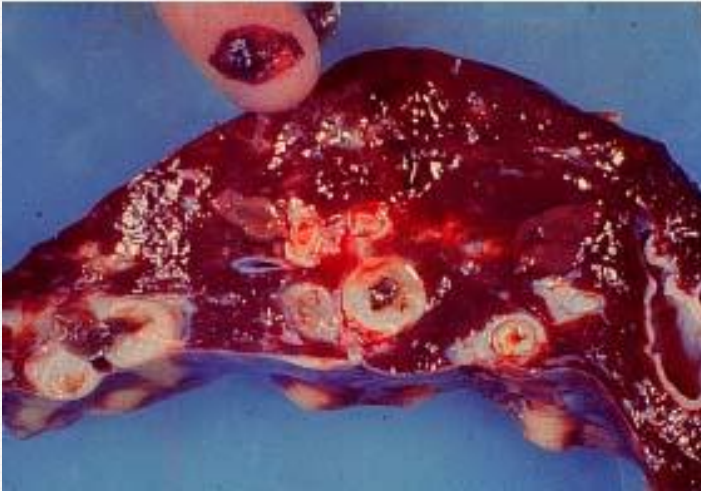
Toxocara/ascaris serology: negative

Fascioliasis (and other flukes)



The Life Cycle of *Fasciola hepatica*.





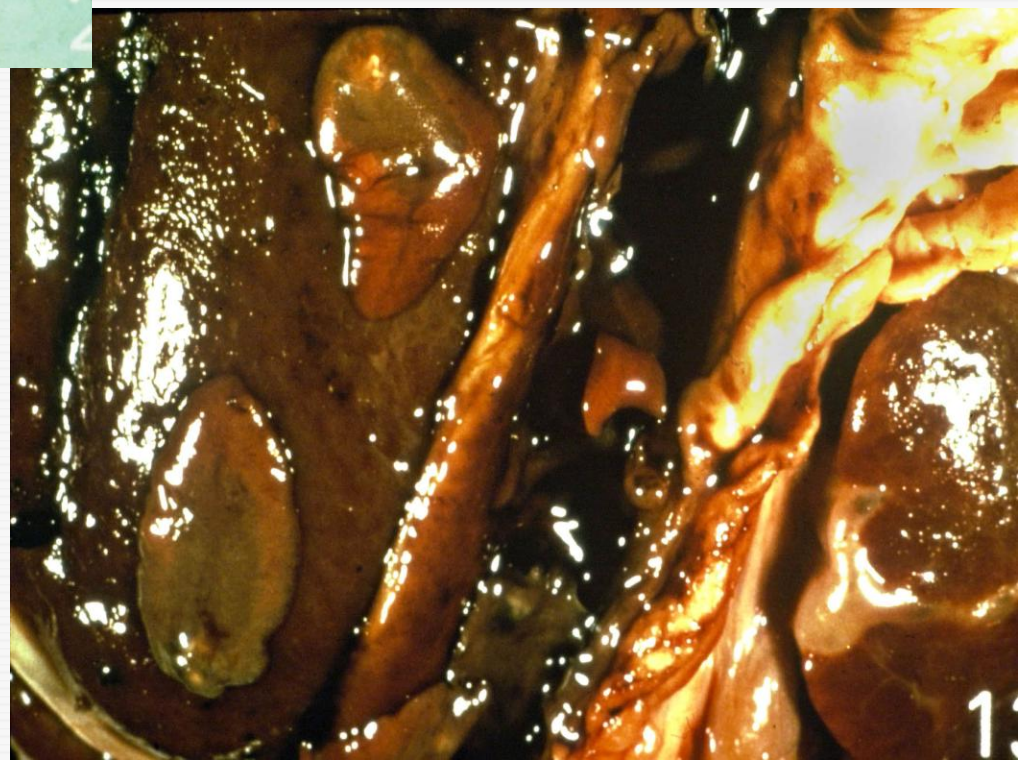
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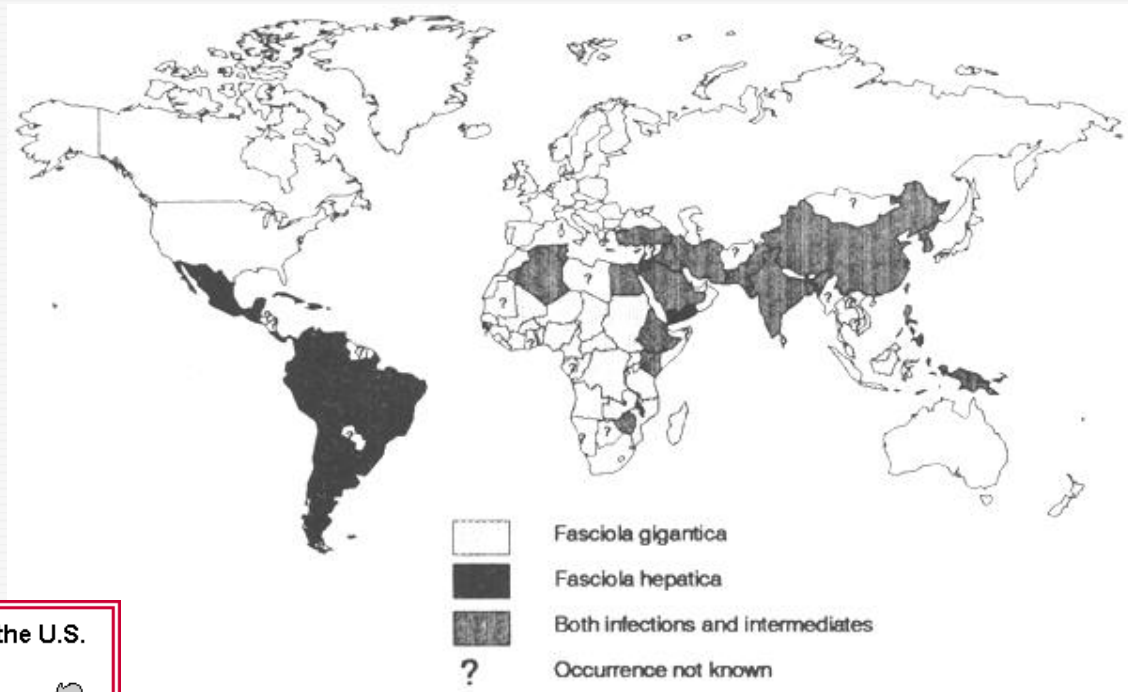
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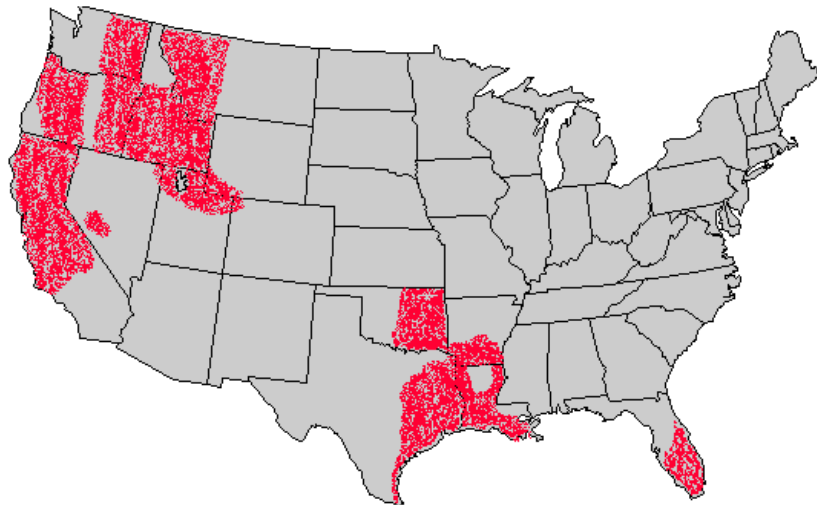
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Fasciola hepatica, global distribution



Approximate Geographic Distribution of *Fasciola hepatica* in the U.S.



Fascioliasis, treatment

- Triclabendazol (10mg/kg)



Mrs. R

Laboratory examination (cont.)

Fasciola hepatica serology: negative

Stool: no eggs

Ultrasound/CT liver: normal

Mrs. R

Laboratory examination (cont.)

IgE: 984 kU/L (N<100)

Phadiatop[®]: negative

Eosinophilia $> 500.10^6/L$

Stool microscopy

pos

Treat

neg

Schistosoma/Filaria/Strongyloides
ELISA (if appropriate)

pos

Further diagnostics

neg

Toxocara/Ascaris serology
ELISA (if appropriate)

pos

Repeat stool/treat

neg

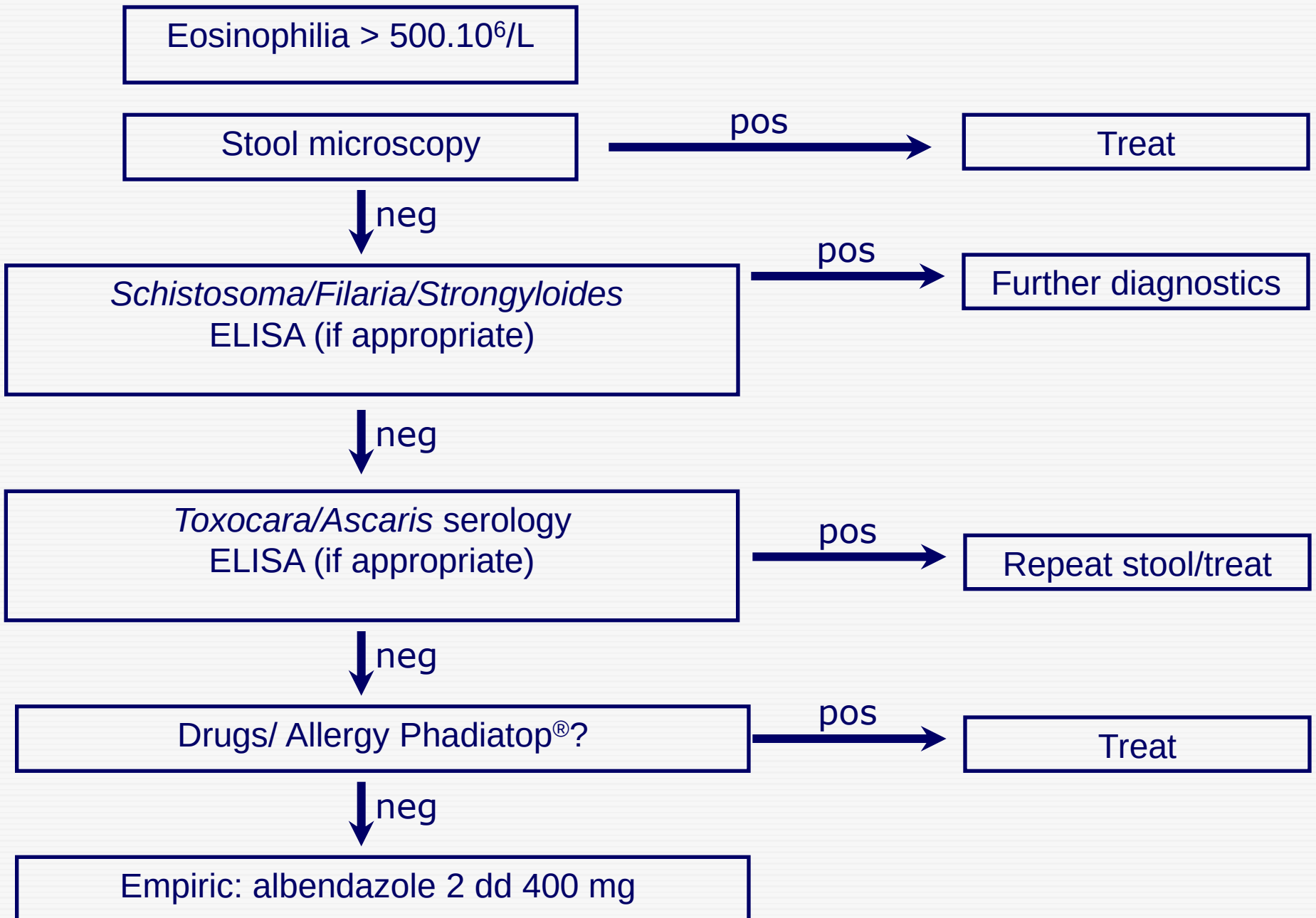
Drugs/ Allergy Phadiatop®?

pos

Treat

neg

Empiric: albendazole 2 dd 400 mg



The end